Evidence-Unit Compression for Financial RAG

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Abstract

Retrieval augmented generation (RAG) can retrieve relevant context and help an answer generator produce better answers. For financial question answering, it retrieves context containing financial values like the periods and units. However, the retrieved context to an answer generator may be too long and redundant. To shorten it, I introduce an evidence-unit compression method especially for financial RAG. This method constructs linked source text and table units, reconstructs dependent context passages when needed, and selects relevant question evidence under a fixed token budget. I evaluate the method on retrieved SEC 10-K filings and the contexts provided by FinQA. On SEC filings, my method increases evidence recall, term retention and answer coverage compared with LLMLingua-2 and LongLLMLingua. On FinQA, it outperforms both compression baselines in program validity and execution accuracy. My findings indicate that this evidence-unit compression preserves more relevant financial context, keeps needed relationships among evidence, and improves question answering under a constrained token budget.

1 Introduction

Retrieval augmented generation (RAG) can provide language models with context for answering questions (Lewis et al., 2020). However, in financial question answering, a retrieved fact requires more detailed domain information, such as the company, unit, or scale. Moreover, it also requires connecting a financial change to an explanation in nearby sentences. As a result, the context becomes too long and redundant.

When a large language model(LLM) is used as an answer generator, the long prompt with redundant information can weaken its performance (Shi et al., 2023). It also brings higher computational costs. In this condition, prompt compression is useful to reduce the amount of context passed to an LLM. For example, LLMLingua-2 is a task-agnostic prompt compression, which can generalize on different tasks and lower the computational cost and latency (Pan et al., 2024). LongLLMLingua works on long-context settings, which can compress well for long prompt with sparse key information (Jiang et al., 2024). However, these methods do not focus on financial area, so they may lose the key domain information.

To address these challenges, the compression method needs to shorten the retrieved financial context under a token budget, but also retain complete and answer-relevant evidence.

Inspired by these requirements, I propose an evidence-unit compression method, which is placed between context acquisition and generation. This compressor first identifies whether a passage needs nearby source context to make its meaning clear. When additional context is needed, it chooses units based on the observed structure of financial context and reconstructs them into self-contained units. Then it uses financial and structural cues to score these units against the question, and select them under a token budget. During this process, it preserves the source metadata associated with each unit.

I evaluate this method in two settings. In the SEC setting, context consists of retrieved chunks from annual 10-K filings. The FinQA setting instead supplies report-local text and tables with each question, and I convert these materials into context units. Together, the two settings test the compressor on retrieved filing passages and provided table–text evidence. I compare against uncompressed context and LLMLingua-2 and LongLLMLingua as two compression baselines under the same token budgets.

On SEC blind-test questions, my method retains more labelled evidence tokens than LLMLingua-2, and increases the answer-term coverage. On the FinQA public-test questions, this compressor also

increases execution accuracy and nearly reaches the performance of the uncompressed condition. These results show that my method outperforms LLMLingua-2 and LongLLMLingua on evidence recall, term retention and answer coverage. It depends on preserving financial evidence as interpretable units. My analysis further examines how evidence retention relates to reasoning and source attribution.

The main contributions of this work are threefold:

- I introduce a structure-adaptive compressor that constructs financial evidence units and selects them under a hard token budget.
- I evaluate the compressor on both the SEC questions and FinQA numerical questions. SEC tests whether the method keeps the evidence needed to answer questions in retrieved filing passages, while FinQA tests whether it keeps the numbers needed to calculate answers from the tables and text provided with each question.
- I analyze finance-aware selection, numerical reasoning depth, and SEC question types to identify where retained evidence contributes to generated answers.

2 Related Work

2.1 Financial Document Question Answering

Financial document question answering (QA) combines ambiguous evidence identification with numerical reasoning over heterogeneous sources. Unlike general-domain QA which mainly considers textual evidence (Yang et al., 2018), financial QA also requires table understanding and long-document evidence retrieval.

Existing financial QA benchmarks have studied these challenges under different context settings. FinQA (Chen et al., 2021) includes expert-written QA pairs over financial reports, providing information from both tables and unstructured texts to answer. TAT-QA (Zhu et al., 2021) uses a combination of tabular and textual content in financial reports, testing the capability of complex reasoning. Convfinqa (Chen et al., 2022) extends this setting to multi-turn conversations, testing long-range, complex numerical reasoning paths in real-world conversations. These benchmarks evaluate reasoning over report-local evidence which has already been associated with each question.

Recent benchmarks broaden the evidence setting. DocFinQA associates FinQA questions with full reports, increasing the average context length (Reddy et al., 2024). FinanceBench (Islam et al., 2023) evaluates open-book financial QA using questions paired with answer and evidence annotations, showing the importance of evidence localization and factual grounding. SEC-QA (Lai et al., 2025) proposes a semi-automatic framework to construct questions requiring evidence from multiple financial documents, which can better represent real-world financial conditions. These benchmarks extend financial QA from reasoning over given evidence to the joint problem of retrieval, evidence selection, and numerical reasoning.

My experiment considers two complementary settings: retrieved passages from SEC Form 10-K filings and table–text context provided by FinQA. This pairing tests whether the same compression method preserves evidence in both retrieved long-document context and compact numerical-reasoning context.

2.2 RAG for Financial Document

Retrieval-augmented generation combines parametric knowledge of a language model and non-parametric memory for language generation (Lewis et al., 2020). Given a question, a RAG system retrieves relevant documents and provides them to a language model for answer generation. In recent years, it has been a viable solution for LLMs when dealing with domain-specific queries, helping the model access information which is not stored in its parameters.

In finance, RAG also plays an important role in retrieving domain knowledge and relevant information in QA workflows. Recent RAG systems have explored graph-based retrieval (Barry et al., 2025) to improve the efficiency depending on knowledge graph design, and multi-aspect retrieval over filings (Wang et al., 2025) for multiple finance reports. It helps unify different data formats and generate metadata summary chunks.

From evidence perspective, HC-RAG (Chen et al., 2026) applies evidence-centric retrieval across heterogeneous filings through document-section-unit paths. Other work improves the evidence supplied to the generator at earlier stages of the pipeline. Yepes et al. (2024) works on financial document chunking, studying structure-aware chunking for financial reports. Choe et al. (2025) combines hierarchical retrieval with evidence cura-

tion over SEC documents.

Unlike prior work that improves document chunking or retrieval, my work examines the subsequent step. I study how to compress this context under a fixed token budget while preserving complete evidence units.

2.3 Prompt and Context Compression

Rich and long-context prompts challenge LLMs for higher computational cost, performance reduction and position bias (Jiang et al., 2024). Prompt compression is a powerful solution to address these problems, reducing the context presented to a language model.

Several methods have been proposed for prompt and context compression. LLMLingua uses a smaller language model to identify less informative tokens under a given compression budget (Jiang et al., 2023). LongLLMLingua extends this method with question-aware compression and document reordering for long-context tasks (Jiang et al., 2024). LLMLingua-2 formulates task-agnostic prompt compression as token classification (Pan et al., 2024), so it supports compression without relying on a specific task.

Other work researches compression specifically for retrieval-augmented language models. RECOMP compresses retrieved documents into textual summaries to improve in-context retrieval language models (Xu et al., 2023). It trains extractive and abstractive compressors to produce a shorter context from retrieved documents (**?**).

Unlike token-level compression, my method selects and reconstructs complete evidence units. It uses query and domain-specific signals to preserve relations between financial values, units, table headers and so on.

3 Method

In this section, I introduce an evidence-unit compression method for financial question answering. First, I introduce the task formulation to understand evidence-unit compression better. Second, I describe the overall pipeline, including the two context-acquisition settings and the subsequent compression and evidence-selection stages. Figure 1 summarizes the complete pipeline.

3.1 Task Formulation

Post-Retrieval Context Compression Given a financial question q, a context acquisition component returns an ordered collection of context units

$$D_q = \langle d_1, d_2, \ldots, d_k \rangle, \tag{1}$$

where each $d_i = (x_i, m_i)$ contains tabular or textual content x_i and its source metadata m_i. The retrieved context may contain both relevant evidence and redundant information, which needs to be shortened while preserving the necessary information.

The post-retrieval compression task is to construct a compressed context $\widetilde{D}_q$ from D_q after the retrieval stage:

$$\widetilde{D}_q = C(q, D_q, B_q), \tag{2}$$

where C is the compression function and B_q is the token budget for question q.

Let $\tau(\cdot)$ denote the tokenizer used to measure context length. The compressed output should satisfy the hard budget constraint

$$\left|\tau\left(\widetilde{D}_q\right)\right| \leq B_q. \tag{3}$$

Evidence Units I treat the retrieved context as a collection of candidate evidence units $\mathcal{U}_q = \{u_1, \ldots, u_n\}$.

An evidence unit is a self-contained range that preserves the information needed to explain a financial fact. It includes an entity, a financial concept, a fiscal period, a unit or scale and the relation among them. For example, a table value should remain the connection of its row label, column year and scale. Similarly, the context about a financial change should preserve both the reported change and its stated cause.

Some retrieved units already provide sufficient context and can be retained directly. Other units can contain fragmented evidence or references that depend on nearby sentences. In such cases, the compressor selects related sentences or table blocks from the same retrieved source, and groups them into a compact evidence package. Within the token budget, it prefers units that are relevant to the question and avoids repeated content.

3.2 Pipeline Overview

I describe the pipeline of this financial QA RAG as shown in Figure 1. This pipeline comprises three main components:

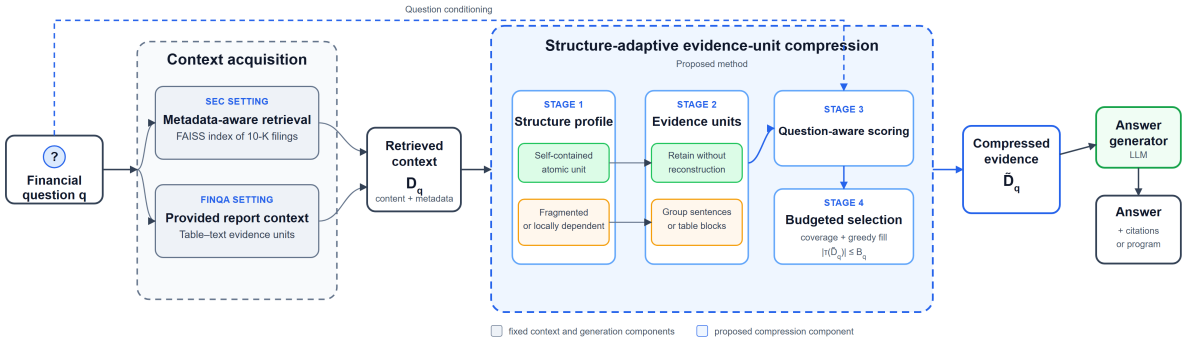


Figure 1: Overview of the financial question answering pipeline

Context Acquisition I evaluate the pipeline under two context-acquisition settings. For SEC questions, I retrieve candidate chunks from filings using metadata-aware search. For FinQA questions, I use the report-local text and table provided with each example. Both settings produce source-linked context units for the compressor.

Evidence-unit Compression The compressor identifies the structure of each context unit and extracts textual or tabular evidence. It retains self-contained ranges directly and reconstructs context-dependent spans with nearby content from the same source. This construction keeps financial values together with the labels, units, and source information needed to interpret them.

Budget-constrained Evidence Selection this part scores the resulting units with respect to the question and selects relevant evidence under a fixed token budget B_q.

The selected evidence units are assembled in source order with their metadata, to form the compressed context $\widetilde{D}_q$. The generator then uses this context to produce a source-cited answer for SEC or an executable numerical program and its answer for FinQA.

3.3 Context Acquisition

Given a financial question q, the context acquisition component constructs the candidate context D_q. As the SEC and FinQA settings provide context in different ways, I use a setting-specific acquisition procedure. During this procedure, each unit $d_i = (x_i, m_i)$ retains its content x_i, source identity, and available metadata m_i, is then passed to the compression component.

SEC Index Construction and Retrieval For building the dense FAISS index, I extract Item 7 from each SEC 10-K filing and divide the document into structure-aware chunks. In annual financial reports, the context provides important details like company name, section title and fiscal year. Each chunk retains this information and uses metadata to store it. The chunking procedure follows section boundaries and keeps consecutive table elements together. When a text or table block exceeds the maximum chunk length, I recursively divide it. I build a dense FAISS index over these chunks (Johnson et al., 2019).

At the retrieval stage, I use a metadata-aware retriever to search a FAISS index and return a ranked collection of document chunks. In detail, company and year mentions in the question are used as metadata filters. The retriever first searches chunks from filings that match these metadata values, and ranks them by their semantic similarity to the question. If no company or year can be found, retrieval is performed over the full index.

For a complex question, the retriever generates a set of focused subqueries. Each subquery returns a ranked list of candidate chunks. I then take one chunk from each ranked list at a time and combine the selected chunks into a single list, so that each subquery contributes evidence to the final context. If the same chunk is retrieved by multiple subqueries, it is included only once. This combined list forms the candidate context D_q.

FinQA Context Construction FinQA (Chen et al., 2021) associates each question with a report-local context that contains pre-table text, a financial table, and post-table text. Since this context is provided by the benchmark, I do not perform retrieval for FinQA. Instead, I convert the provided content into the same context-unit representation used in the SEC setting.

For the textual content, each pre- and post-table

text segment is treated as a separate unit. For the tabular content, each row is converted into a table unit, which links its row label to the corresponding column headers and cell values. The units remain in their source order and carry metadata identifying the report, example and unit type. They together form the candidate context D_q.

3.4 Evidence-Unit Compression

After context acquisition, the final context D_q may be too long and contain overlapping information. Also, it may contain passages whose meaning depends on nearby sentences or table headers. Based on this situation, I propose the evidence-unit compression. It contains two steps: source structure identification and evidence-unit construction.

Identifying Source Structure For each source in D_q, the compressor first determines how its evidence units should be formed. It examines the source's structure, including unit type, token length, number of paragraphs and references to earlier content. These features help determine whether the extracted units can be used on their own or need additional context.

Evidence-Unit Construction The compressor then extracts candidate units from each source. It divides text passages into sentences, and combines tabular fragments into short blocks. For sources with clear unit boundaries, the compressor directly forms candidate units. When a source depends on nearby context, the compressor reconstructs it with relevant sentences and tabular content. For example, if a unit begins with "this increases...", it is joined with the last unit from the same source. After construction, each final unit remains linked to its source and original position. These units form the candidate set $\mathcal{U}_q$.

3.5 Budget-Constrained Evidence Selection

Given the candidate units $\mathcal{U}_q$, the compressor determines which units to include in the final context. It first scores the relevance of units to the question, and then selects them under the token budget B_q.

Question-Aware Scoring Based on the question and the unit's source, the compressor gives each unit a score. This score reflects matches to the question's key terms and fiscal year, also numerical values and units. When the question asks about a cause or a change, units that contain relevant expressions will receive additional weight. A section title that matches the question and an earlier retrieval rank receives a smaller score increase. By contrast, routine filing phrases such as a table of contents receive a penalty. For sources requiring further processing, question terms that appear in many retrieved sources contribute less to the score.

Budgeted Selection After scoring, the compressor first tries to retain evidence from different sources. When the context contains sources which require further processing, the compressor attempts to retain a high-scoring unit from each source group. This type of group is defined by the company and fiscal year in the source metadata.

Next, it tries to retain units matching key terms in the question. If all sources are identified as self-contained, the compressor starts by selecting units that match key terms in the question. Next, the compressor considers the remaining units, giving lower priority to units that repeat already selected information. A unit is added only if the assembled context remains within the token budget B_q. Finally, the selected units are assembled by source in their original order, with their source metadata retained. These units form the compressed context $\widetilde{D}_q$, which is passed to the answer generator.

4 Experiment

In this section, I evaluate whether my compression method preserves necessary financial evidence under a limited context budget, and whether the retained evidence supports a better answer generation.

4.1 Datasets

I apply the compression method to two dataset settings. The SEC experiment evaluates compression after document retrieval, and FinQA tests the method on report-local context provided with each question.

SEC Filings I build an evaluation set from the Item 7 sections of 100 annual 10-K filings, covering 20 companies. The filings are divided into 6,889 chunks and indexed together. I construct 260 questions split into development, validation, and blind-test sets containing 100, 60 and 100 questions separately.

In the experiment, I use the development set to design the method, the validation set to choose between single-query and multi-query retrieval, and the blind-test set for the final comparison.

FinQA I also evaluate on the 1,147 questions in the FinQA public test set (Chen et al., 2021). In this test set, each question comes with report-local text and a table. To let compressor choose which parts to keep, I divide these text and table into context units before compression. As FinQA does not involve retrieval from the SEC filing index, it tests whether the same compression method works with a different form of financial evidence.

4.2 Baselines

I compare my method with an uncompressed baseline and two prompt compression methods. All conditions use the same candidate context. Within each dataset, I use the same answer generator and prompt.

Uncompressed This condition passes the full candidate context to the answer generator without compression. I use it as a reference for changes in evidence preservation and answer quality after compression.

LLMLingua-2 LLMLingua-2 is a task-agnostic compression method (Pan et al., 2024). It uses data distillation to learn from an LLM and treats prompt compression as token classification, selecting which tokens to keep from the original prompt. Because it is designed to work across different tasks, I use it to measure the performance difference between my method and a general-task compressor.

LongLLMLingua Different from LLMLingua-2, LongLLMLingua is a question-aware compression method(Jiang et al., 2024). It is designed to compress prompts in long context settings. My method is also to compress long context prompt in financial scenarios, which is similar. So the comparison provides a more direct test of my method in that setting.

4.3 Implementation Details

For SEC retrieval, I use text-embedding-3-small to encode filing chunks and store the embeddings in a FAISS index.

I keep at most ten chunks per question. When query expansion is used, I generate no more than four subqueries in addition to the original question and retrieve up to three candidates per query. On the validation set and blind-test evaluation, conditional bounded MultiQuery was selected over single-query retrieval. Appendix A reports the configuration comparison and additional experimental details.

I use llmlingua version 0.2.2 for both compression baselines. LLMLingua-2 uses the multilingual BERT MeetingBank checkpoint, with a target retention rate of 0.575. LongLLMLingua uses the Phi-2 checkpoint and compresses each source separately.

For the context length, I measure it with cl100k_base tokenizer. I set LLMLingua-2's actual output context length as the token limit. It applies for my method per question and for LongLLMLingua per source. At the budget-constrained evidence selection stage, I also use limited context length to check if the evidence can be a complete unit. This check includes length limits of 180 tokens for untyped text, 520 for other units with clear boundaries, and 720 for table units.

For answer generation, all conditions use GPT-4.1-mini with temperature 0 and the same prompt within each dataset. The FinQA prompt includes two manually written examples, which do not include any benchmark question or answer.

4.4 Evaluation Metrics

I evaluate the pipeline at three levels: retrieval, compressed context, and generated answers. Retrieval metrics apply only to SEC, while the answer-level metrics differ between SEC and FinQA.

Retrieval Metrics. For SEC, I report Hit@1, Recall@10, and mean reciprocal rank (MRR) on financial questions. Hit@1 measures whether the first retrieved chunk is labelled as relevant. Recall@10 measures the fraction of labelled relevant chunks found among the top ten results. MRR uses the rank of the first relevant chunk.

Compression Metrics. To measure compression, I use context retention, evidence recall, expected-term retention for SEC and operand recall for FinQA. Context retention is the number of output context tokens divided by the number of input context tokens. Evidence recall is the fraction of distinct, normalized tokens from the labelled useful evidence in compressed context. Expected-term retention for SEC measures the fraction of annotated answer terms kept in the context. Operand recall for FinQA applies the same calculation, measuring the fraction of values used in the reference that remain in the compressed context.

SEC Answer Metrics. I set several deterministic measures of answer content for SEC, such as answer coverage, labelled citation, and corrected pass. Answer coverage is the fraction of annotated answer terms appearing in the generated output. Labelled citation is the proportion of answers that cite at least one labelled relevant chunk. Corrected pass applies a rule to each question: an answerable response must have valid citations and cite labelled evidence, while an unanswerable response must refuse to answer.

FinQA Answer Metrics. For FinQA, I use different answer metrics, including valid program and execution accuracy. Valid program is the proportion of generated programs that can be parsed and executed. Execution accuracy is the proportion whose executed result matches the reference answer. Thus, execution accuracy evaluates the numerical result.

Statistical Analysis. For the main comparisons, I evaluate methods on the same questions and report their paired score differences. I estimate 95% confidence intervals by resampling questions with replacement. In this estimation, I use 5,000 resamples for the SEC comparisons, and 10,000 for FinQA part. This analysis shows whether the observed differences remain similar when the evaluation questions change.

5 Results

In this section, I first report results on the SEC blind test and the FinQA public test. Then I examine paired differences among all four methods. The SEC results measure the effect of compression after retrieval, whereas FinQA evaluates compression of the report context supplied with each question.

5.1 Results on SEC Filings

Before the compression, the retriever achieves 43.16% Hit@1, 67.89% Recall@10, and 0.6005 MRR on answerable blind-test questions. During the compression, I use the same retrieved chunks to keep it comparable. Table 1 compares their effects on the compressed context and generated answers.

At nearly the same retention as LLMLingua-2, my method preserves more labelled evidence (88.23% versus 68.32%) and expected answer terms (62.81% versus 21.93%). Its answer coverage of generated answer is also higher and approaches the uncompressed result. Compared with LongLLMLingua, my method achieves higher evidence recall (88.23

However, the improvement in evidence preservation does not happen on every answer-level measure. My method obtains 83.16% labelled citation and 83.00% corrected pass, below the corresponding numbers of the other methods in Table 1. It shows that retaining relevant content does not always lead the generator to cite the labelled source.

5.2 Results on FinQA

Table 2 reports results on all 1,147 public-test questions. At 58.27% context retention, my method preserves 91.97% of labelled evidence tokens and 94.78% operand recall. These values exceed those of both compression baselines.

For the generated programs, my method's execution accuracy is 26.42%, compared with 23.02% for LLMLingua-2 and 12.21% for LongLLMLingua. The uncompressed baseline remains higher in execution accuracy at 27.90%. However, my method achieves the highest valid-program rate, at 89.01%. Nevertheless, the gap between program validity and execution accuracy shows that a program can be executable without producing the correct result.

5.3 Paired Comparisons and Trade-offs

Paired comparisons show whether the differences in the tables are stable across different samples of questions. On SEC, my method improves evidence recall over LLMLingua-2 by 19.91 percentage points (95% CI:[17.17, 22.56]) and answer coverage by 7.37 points([2.46, 12.28]). Relative to LongLLMLingua, its answer coverage is 17.72 points higher ([12.63, 22.98]). However, its corrected pass rate is 4.00 points lower than LongLLMLingua's.

On FinQA, my method improves execution accuracy over LLMLingua-2 by 3.40 points ([1.39, 5.41]) and over LongLLMLingua by 14.21 points ([11.86, 16.56]). Compared with the uncompressed context, it is only 1.48 points lower. It indicates that the method retains most of the original performance while using a compressed context.

Together, these results support better evidence preservation and recovery of performance lost under the compression baselines. However, better information preservation does not necessarily lead to a higher corrected pass rate.

6 Analysis

In this part, I examine the contribution of financial selection features and the effect of reasoning depth

Method	Compressed context			Generated answer		
	Retention	Evidence recall	Term retention	Answer coverage	Labelled citation	Corrected pass
Uncompressed	100.00	–	–	51.75	86.32	87.00
LLMLingua-2	58.10	68.32	21.93	43.68	86.32	86.00
LongLLMLingua†	56.92	80.34	27.37	33.33	86.32	87.00
My method	58.10	88.23	62.81	51.05	83.16	83.00

Table 1: SEC blind-test results (%). Retention, evidence recall, term retention, answer coverage, and labelled citation are reported on the 95 answerable questions; corrected pass includes all 100 questions.

Method	Compressed context			Generated program	
	Retention	Evidence recall	Operand recall	Valid program	Execution accuracy
Uncompressed	100.00	94.90	98.18	87.36	27.90
LLMLingua-2	58.58	68.59	60.08	78.12	23.02
LongLLMLingua†	58.32	80.79	49.95	71.75	12.21
My method	58.27	91.97	94.78	89.01	26.42

Table 2: FinQA public-test results on 1,147 questions (%). Retention is computed from total output and input context token, and operand recall is defined for 1,146 questions.

Condition	Tokens	Evi. R	Oper. R	Exec.
Full method	87,316	92.74	87.32	50.71
Atomic + query	86,806	92.49	78.81	47.14

Table 3: Ablation on 140 FinQA development questions. Evi. R is labelled-evidence token recall, Oper. R is exact operand recall, and Exec. is execution accuracy. All three are percentages.

Method	Single-step ($n = 654$)	Multi-step ($n = 493$)
Uncompressed	36.39	16.63
LLMLingua-2	30.28	13.39
LongLLMLingua	15.29	8.11
My method	35.17	14.81

Table 4: FinQA public-test execution accuracy by reasoning depth(%).

on FinQA. Besides that, I analyze the relationship between retrieval and source attribution in SEC questions.

6.1 Ablation Study

To assess the contribution of financial cues for selecting evidence, I compare the full method with an atomic-and-query variant on 140 FinQA development questions. It's a version that selects complete evidence units mainly by their overlap with the question. Both variants extract atomic evidence units and use the same per-question token caps. The atomic-and-query variant removes the year, numerical, relational, section, and table scoring signals. Table 3 reports their realized token counts and performance.

Table 3 shows similar evidence-token recall for the two variants (92.49% versus 92.74%), alongside an increase in exact operand recall from 78.81% to 87.32% with the full method. The paired gain is 8.51 percentage points. Execution accuracy changes from 47.14% to 50.71%. This pattern highlights the contribution of financial and structural scoring signals to selecting relevant values from otherwise relevant evidence.

6.2 Numerical Reasoning

All 1,147 FinQA public-test questions follow the atomic-selection path. To examine how performance varies with calculation complexity, I separate questions by the number of reasoning steps. Table 4 reports execution accuracy for each group.

My method exceeds LLMLingua-2 by 4.89 points on single-step questions and 1.42 points on multi-step questions. Accuracy decreases with reasoning depth for every condition, and even uncompressed context only reaches 16.63% on multi-step questions.

The measures further distinguish evidence availability from successful calculation. The main results in Table 2 show that my method retains most values needed for the reference calculations. However, numerical reasoning shows that execution

accuracy is always lower on multi-step questions across all conditions. These results together identify that even when much of the numerical evidence is preserved, the multi-step calculation is still an important challenge.

6.3 Evidence Retrieval and Source Attribution

I analyze SEC performance by question type to distinguish the retrieval of labelled evidence in generated answers. For comparison, all three compression methods receive the same retrieved chunks. Table 5 reports retrieval recall, answer-term coverage, and labelled-evidence citation on the 95 answerable blind-test questions.

For answer-term coverage, table 5 shows that my method achieves the highest answer-term coverage on single-section, cross-section, and cross-year questions. The largest gain over LLMLingua-2 occurs on cross-year questions, where coverage rises from 54.00% to 66.67%, and LongLLMLingua reaches 44.67%. Cross-company questions have the lowest Recall@10 at 54.00%. For this type of question, my method and LLMLingua-2 both reach 40.67% answer-term coverage, compared with 29.33% for LongLLMLingua.

Labelled citation is the same for all three methods on single-section and cross-section questions. On cross-year questions, my method reaches 80.00%, compared with 84.00% for both baselines; on cross-company questions, the corresponding values are 76.00% and 84.00%. These results show how retrieving relevant chunks, retaining answer terms, and citing labelled sources each contribute to the grounded SEC responses.

I further examine retrieval order through the candidate-preserving graph reranker described in Appendix B. With the Top-10 candidate set unchanged, reranking raises blind-test Hit@1 from 43.16% to 55.79%. For my compressor, labelled citation rises from 83.16% to 85.26%, while answer-term coverage changes from 51.05% to 47.19%. This comparison shows that moving labelled evidence earlier can change both source attribution and the answer terms used in the response.

7 Conclusion

This research presents an evidence-unit compressor for financial question answering under a fixed token budget. The method constructs text and table units with the context needed to interpret financial facts, then selects question-relevant units while retaining their source information. On retrieved SEC contexts and FinQA's provided table–text contexts, it improves evidence retention over LLMLingua-2 and LongLLMLingua at nearly 58% context retention. On FinQA, my method achieves higher execution accuracy than LLMLingua-2 and LongLLMLingua. These experimental results support evidence-unit selection as a way to reduce the loss of useful financial context during compression. However, improved evidence retention does not always guarantee better evidence use. The labelled-source citation does not improve on SEC, and multi-step numerical reasoning remains difficult on FinQA.

Limitations

The current work still has some limitations. Firstly, the evaluation covers two specific forms of financial evidence, so the performance on other filing sections and document types remains untested. The SEC corpus contains Item 7 passages from 100 Form 10-K filings by 20 companies. Although the companies used for blind-test questions do not appear in the development or validation sets, their filings are already in the retrieval index. As a result, this test does not measure performance when new filings are added to the index.

Secondly, the SEC questions and supporting-evidence labels were machine-curated and only passed deterministic consistency checks. They were not independently reviewed for financial-semantic correctness.

Finally, preserving evidence does not eliminate challenges at the answer-generation stage. On the SEC blind test, my method has lower labelled citation and corrected pass rates than the other compression baselines. In addition, multi-step FinQA questions remain difficult even with uncompressed context.

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			Answer-term coverage			Labelled citation		
Question type	n	Recall@10	LLM-2	Long	My	LLM-2	Long	My
Single-section	20	85.00	52.50	36.67	**61.67**	85.00	85.00	85.00
Cross-section	25	64.00	29.33	23.33	**37.33**	92.00	92.00	92.00
Cross-year	25	72.00	54.00	44.67	**66.67**	84.00	84.00	80.00
Cross-company	25	54.00	**40.67**	29.33	**40.67**	84.00	84.00	76.00

Table 5: SEC blind-test results by answerable question type (%). Recall@10 is the fraction of labelled relevant chunks found among the top ten retrieved chunks. Answer-term coverage and labelled citation are measured on generated responses. LLM-2 denotes LLMLingua-2; Long denotes LongLLMLingua.

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A Additional Experiment Details

SEC benchmark construction. The SEC corpus contains 6,889 chunks from the Item 7 sections of 100 annual 10-K filings. I partition the 20 companies into development, validation, and blind-test groups: 10 companies and 100 questions for development, 5 companies and 60 questions for validation, and 5 companies and 100 questions for blind testing. The corresponding numbers of answerable questions are 95, 55, and 95.

The validation and blind-test sets contain single-section, cross-section, cross-year, and cross-company questions. Each type contributes 15 questions to the validation set and 25 questions to the blind-test set. Of these, five questions in each split are unanswerable controls. For each answerable question, the annotation records the chunks containing the relevant evidence. It also includes supporting passages, a reference answer, and expected answer terms.

Retrieval condition	Hit@1	Recall@10	MRR
Single query	40.00	58.18	0.5485
Conditional MultiQuery	43.64	**73.64**	**0.5776**

Table 6: SEC retrieval on the 55 answerable validation questions. Hit@1 and Recall@10 are percentages; higher values are better.

Questions and annotation candidates are generated from selected filing passages.They are then checked for structural consistency and exact matches with the source text.

Retrieval configuration. I made one bounded retrieval choice on the SEC validation set. The single-query and conditional multiquery settings use the same FAISS index and return at most ten chunks. Single-query retrieval directly uses the original question. Conditional multiquery first decomposes a complex question into at most four additional queries while keeping the original question. Next, it retrieves at most three candidates for each query, and merges the results after removing repeated chunks. The query generator can only see the questions to ensure a fair comparison. Table 6 reports the validation comparison.

FinQA context and answer generation. All FinQA conditions use GPT-4.1-mini at temperature zero and same fixed output format. For each question, the model returns a JSON object containing a sequence of calculation steps, a final answer, and source citations. The calculation steps use the operations defined in FinQA, including arithmetic and calculations over table values. When a later step depends on an earlier result, it refers to that result by its step number.

I use two manually written examples to show the model how to follow this output format. One demonstrates a percentage-change calculation, and the other demonstrates an average over table values. The two examples use fictional companies and numbers which do not belong to any FinQA benchmark items.

After generation, I execute the predicted calculation steps to obtain the final numerical result. Then I compare this result with the reference answer for evaluation.

Budget matching. LLMLingua-2 uses a target retention rate of 0.575. Its output token count sets the maximum number of context tokens available to my method for each question. For LongLLM-

Lingua, the LLMLingua-2 output token count for each source sets the corresponding token limit.

B Exploratory Graph-Based Reranking

Graph construction and controlled comparison. I also examined whether filing structure can improve the ordering of the existing SEC retrieval candidates. I constructed a property graph from the same 6,889 chunks. The graph uses filing metadata, section titles, explicit years, and a fixed set of financial-concept aliases. It contains 10,452 nodes and 44,239 edges. The nodes represent companies, filings, fiscal years, sections, chunks, and financial concepts. The edges represent relations such as filing ownership and section and chunk containment.

For each question, the graph ranking uses matched companies, years, concepts and section terms. In the main graph condition, I use reciprocal rank fusion with equal weights to combine the graph ranking and the vector ranking used in the main experiment (Cormack et al., 2009). The combined ranking is restricted to the original vector Top-10, so a chunk outside this set cannot enter the final context. As a result, the two conditions contain the same Top-10 chunks for every question and differ only in their order. I also evaluate graph-only retrieval and retrieval over the unrestricted union of graph and vector candidates as diagnostic conditions.

Retrieval effects. Table 7 compares the two rankings on answerable SEC questions.

Split	Ranking	Hit@1	Recall@10	MRR
Val. ($n = 55$)	Vector	43.64	73.64	0.5776
	Graph	**60.00**	73.64	**0.7142**
Blind ($n = 95$)	Vector	43.16	67.89	0.6005
	Graph	**55.79**	67.89	**0.6883**

Table 7: SEC retrieval on validation (Val.) and blind-test (Blind) answerable questions. Graph reranks the same vector Top-10 candidates. Hit@1 and Recall@10 are percentages, and MRR ranges from 0 to 1. Higher is better.

Graph reranking moves labelled evidence earlier in both splits. On the blind test, Hit@1 increases from 43.16% to 55.79%, and MRR increases from 0.6005 to 0.6883. Recall@10 is unchanged because candidate membership is fixed.

End-to-end effects. I next crossed the two retrieval orders with three context conditions: uncompressed context (A), LLMLingua-2 (B), and my compressor (C). The question set, candidate chunks, answer generator, and scoring procedure are shared across the two retrieval orders. Within each order, C is limited to B's realized per-question token count. Table 8 reports the six blind-test conditions.

Ranking	Cond.	Coverage	Citation	Pass
Vector	A	51.75	86.32	87.00
	B	43.68	86.32	86.00
	C	51.05	83.16	83.00
Graph	A	52.46	86.32	86.00
	B	41.40	85.26	86.00
	C	47.19	85.26	86.00

Table 8: SEC blind-test generation under two orderings of the same Top-10 candidates (%). Coverage is answer-term coverage and Citation is labelled-evidence citation, both on 95 answerable questions. Corrected Pass includes all 100 questions. Higher is better.

Under graph reranking, my method raises answer coverage from 41.40% for LLMLingua-2 to 47.19%, a paired difference of 5.79 percentage points (95% CI: [1.05, 10.70]). Relative to the vector ordering of my method, graph reranking raises labelled citation from 83.16% to 85.26% and corrected pass from 83.00% to 86.00%, while answer coverage changes from 51.05% to 47.19%. The paired coverage difference is -3.86 points (95% CI: $[-7.89, -0.18]$). These comparisons show that filing-structure signals can improve the early ranking of available evidence, while the resulting order also changes which answer terms the compressor and generator ultimately use.